The Selling closure for lattice comparison: Euclidean search keys and exact reindexing operators

Petrus H. Zwart *

Molecular Biophysics and Integrated Bioimaging Division, Lawrence Berkeley National Laboratory, 1 Cyclotron Road, Berkeley, CA 94720, USA.

E-mail: phzwart@lbl.gov

lattice reduction; **Selling reduction**; **obtuse superbase**; **lattice isometry**; **reindexing**; **nearest-neighbour search**; **serial crystallography**; **Protein Data Bank**

Abstract

Every crystallographic unit cell is a choice of basis for a lattice, and a lattice has infinitely many bases. Classical reduction (Niggli, Buerger, Delone) selects one representative cell per lattice, which suits tabulation and lattice typing but not comparison: an arbitrarily small change to a lattice can make its representative jump. Searching archives of hundreds of thousands of cells, following lattices through thousands of serial-crystallography frames, and tracing a lattice along a perturbation trajectory are comparison tasks, and they need a coordinate that does not jump. The distance between two lattices is defined ideally first: the smallest symmetrised change to one lattice's definition that makes it congruent to the other, a minimum over integer basis changes that is attained, zero exactly at congruence, and computable with a termination certificate. The rest of the method is a ladder of certified accelerations of that ideal, and the working stages are two of its rungs. For search, each lattice receives one

vector of six numbers: its Selling scalars divided by the square root of their sum and sorted into increasing order. The key is identical for every basis of the lattice at every Voronoi type, scales as a length, is Lipschitz in the metric tensor where square-root products are not, and is indexed with a standard KD tree; its Euclidean distance is a lower bound on the orbit distance between lattices, so a radius query cannot miss a match. For certification, candidates are compared exactly: the finite set of obtuse superbases of each lattice, here called the Selling closure, is enumerated, and an integer change-of-basis matrix is constructed from retained integer coordinates and verified against the metric tensor; symmetry-related alternatives are returned as a coset rather than lost. Two results are built on these stages. The first is an atlas of the Protein Data Bank: 206 214 deposited cells, volume-normalised, occupy a stratified shape space in which the crystal systems emerge without being inputs; the three cubic lattices are single computable points, the tetragonal and rhombohedral families are curves meeting them at exact nexus points, and distinct families run together along asymptotic seams at extreme axial ratios. The second is reindexing: identifying the lattice first, over the closure, returns the exact integer operator and its full automorphism coset without a symmetry tolerance, recovering basis relations that a 3° angular gate drops and that entry-bounded matrix searches silently miss. Family-scale (hen egg-white lysozyme) and per-crystal (a 12 474-crystal XFEL stream) analyses follow as uses of the same coordinate.

1. Introduction

A lattice is a set of points. A unit cell describes that set by three vectors v_1, v_2, v_3, and the metric tensor $\mathbf{G} = \mathbf{B}^\mathsf{T}\mathbf{B}$ collects their lengths and angles. Any integer matrix $\mathbf{M}$ of determinant ± 1 produces another equally valid cell for the same lattice, $\mathbf{B}' = \mathbf{B}\mathbf{M}$ and $\mathbf{G}' = \mathbf{M}^\mathsf{T}\mathbf{G}\mathbf{M}$, so a lattice is not one metric tensor but an infinite family of them.

The traditional way to remove the ambiguity is reduction: a rule that selects one cell from the family. Niggli reduction (Niggli, 1928; Grosse-Kunstleve *et al.*, 2004) is the standard example, and for its original purposes — reporting a cell, assigning a Bravais type, looking a cell up in a table — it is exactly right. It has one unavoidable side effect. A rule that picks one representative must decide what to do at the boundaries where two representatives are equally good, and at those boundaries the chosen cell changes abruptly while the lattice changes smoothly.[S6] The lattice is continuous; the coordinate assigned to it is not. For classification this rarely matters. For comparison it matters constantly: to ask how close two lattices are across an archive, across the frames of a serial experiment, or along a heating or dehydration series, one needs a coordinate that moves a little when the lattice moves a little.

There is a second, less obvious demand. A comparison coordinate necessarily discards basis information, but reindexing — transforming Miller indices, coordinates or symmetry operators from one cell setting to another — needs that information back, and needs it exactly. A method that compares well but reindexes approximately is of limited use in a crystallographic pipeline.

The proposal made here is to define the distance ideally first, and then to speed it up. Ideally, the distance between two lattices is the smallest symmetrised change to one lattice's definition that makes it congruent to the other — a minimum over integer changes of basis that is attained, is zero exactly at congruence, and is computable with a termination certificate (Section 2.2). Everything else in the paper is a certified acceleration of that ideal, down to a six-number Euclidean key: search is done on an object whose distance is a lower bound on the aligned comparison, and what it retrieves is refined by exact procedures that end, when needed, at the ideal itself — nothing within the search radius is missed, and nothing retrieved is trusted until certified. The ingredients are classical or recent, and the paper is explicit about which is which.

Selling reduction and the obtuse superbase go back to Selling, Delone and Bravais (Bravais, 1850; Delone, 1932), the conorm/vonorm language is that of Conway & Sloane (1992), and the crystallographic S^6 embedding, the uniqueness of the reduced Selling scalars as a list and the practical reduction algorithm are due to Andrews, Bernstein and Sauter (Andrews & Bernstein, 1988, 2014, 2016, 2023; Andrews *et al.*, 2019). The type-dependent enumeration of obtuse superbases, the completeness proof of the resulting isometry classification and the perturbation bound on individual root products are Kurlin's (Bright *et al.*, 2021; Kurlin, 2026). What is added here is elementary: the observation that sorting the six root products gives one key per lattice at every Voronoi type, a lemma that its Euclidean distance is a lower bound on the orbit distance, a noise analysis at symmetric lattices, and an exact reindexing procedure over the complete finite set of obtuse superbases — the *Selling closure* — that returns the automorphism coset without a symmetry tolerance. The paper reports two results built on these pieces: an atlas of the Protein Data Bank in lattice-shape space, organised by a computable skeleton of symmetry strata, and exact reindexing against a reference over the closure. The main text keeps to the constructions and the results; derivations and proofs are in Appendices A–D, and every statement that can be checked by computation carries a mark $^{[\mathrm{S}n]}$ pointing at the corresponding check in the Supporting Information.

2. Theory and methods

2.1. Obtuse superbases and the Selling closure

Selling reduction (Appendix A) turns the infinite family of cells into a finite one. Adding a fourth vector $v_0 = -(v_1 + v_2 + v_3)$ gives a *superbase*, four vectors that sum to zero; reduction adjusts them until every pair meets at a right or obtuse angle. The six *conorms* $p_{ij} = -v_i \cdot v_j \geq 0$ of an obtuse superbase are the negatives of the Selling

scalars of Andrews *et al.* (2019). For a generic lattice the obtuse superbase is unique up to relabelling the four vectors and negating all of them — 24 relabellings, 48 with the sign.[S2]

For symmetric lattices this is no longer true. Kurlin (2026) shows that the obtuse superbase is unique up to relabelling only when the Voronoi cell of the lattice is the generic truncated octahedron (type V1); for the four degenerate types V2–V5, further obtuse superbases exist that are not relabellings of one another, obtained from the reduced one by a Selling move at a conorm that is exactly zero. An orthorhombic P lattice, for example, has 32 obtuse superbases (counted as unordered sets of four vectors, the superbase and its global negation being distinct) falling into four families.[S2] The complete finite set of obtuse superbases of a lattice Λ is called here its *Selling closure* $\mathcal{S}(\Lambda)$; Table 1 gives its size and structure by Voronoi type, and Appendix A gives the rule that reads the type off the zero pattern of the reduced conorms.[S3] Three facts matter for what follows. The closure is finite, so it can be enumerated. It is complete and unique as a set — Kurlin's lemmas guarantee that no obtuse superbase of the lattice exists outside it, and no selection or tie-breaking enters — and this is the fact everything below stands on, because an enumeration that provably misses nothing can prove negatives: that no congruence exists, that no operator was dropped. And the classical statement that the closure consists of "24 relabellings" is correct only for lattices with no symmetry: monoclinic P lattices are V4 and orthorhombic P lattices are V5,[S4] so for a large fraction of real cells the shortcut undercounts.

2.2. The ideal: a certified, symmetrised deformation distance

What should the distance between two lattices be? The requirement adopted here is operational: the smallest perturbation of one lattice's definition that makes the two lattices the same. Three short steps turn the requirement into a formula. First, a lat-

tice's definition is its metric tensor $\mathbf{G}_A$ in the reduced frame it is reported in, so a perturbation is a symmetric matrix $\Delta\mathbf{G}$ added to it. Second, the perturbed lattice is congruent to B precisely when an integer change of basis carries one metric onto the other, $\mathbf{G}_A + \Delta\mathbf{G} = \mathbf{M}^{\mathsf{T}}\mathbf{G}_B\mathbf{M}$ for some integer $\mathbf{M}$ of determinant ± 1; solving for the perturbation shows that the congruence-achieving perturbations form the discrete set $\Delta\mathbf{G} = \mathbf{M}^{\mathsf{T}}\mathbf{G}_B\mathbf{M} - \mathbf{G}_A$, one member for each $\mathbf{M}$. Third, "smallest" means minimum norm over that set, and the minimisation over perturbations has become a minimisation over basis changes:

$$d(A \to B) = \min_{\mathbf{M}} \left\| \mathbf{G}_A - \mathbf{M}^{\mathsf{T}}\mathbf{G}_B\mathbf{M} \right\|, \tag{1}$$

with the reported distance the symmetrised $d(A,B) = \max\{d(A \to B),\, d(B \to A)\}$. Nothing in equation (1) is postulated: the integer group appears because congruence is an integer change of basis, and the minimum because the perturbation was required to be smallest. Two declarations remain — the norm (Frobenius on the metric tensor) and the frame in which $\Delta\mathbf{G}$ is measured (A's reported reduced frame, where the perturbation acts) — and the derivation itself shows how the two lattices enter asymmetrically: B only as a lattice, because re-presenting it maps $\mathbf{M} \mapsto N\mathbf{M}$ and merely reparametrises the set, while A enters through its declared frame, which is what the symmetrised maximum then balances. The set being discrete and locally finite, the minimum is attained and is zero exactly at congruence; the minimiser returns $\Delta\mathbf{G}$, which converts directly into the six cell-parameter changes that deform A into a copy of B. For the face-straddling pair of Appendix D the answer is one angle moved by 0.85° and essentially nothing else, and the distance is exactly the 4δ that physically separates the pair — continuous through the face where the obtuse-restricted comparison is not.[S33]

How can a minimum over an infinite group be computed? The first move is Selling reduction, and what it buys is not a normal form but an enumeration. By Kurlin's

completeness lemmas the closure of Section 2.1 is the set of *all* obtuse superbases of a lattice, finite and unique as a set. If A and B are congruent, the two closures are therefore the same geometric object seen in two coordinate systems: every member of $\mathcal{S}(A)$ equals a member of $\mathcal{S}(B)$ after the change of basis. Matching members across the two closures attains the minimum of equation (1) exactly — zero — and returns not one operator but the entire relating coset, in at most $32 \times 32 \times 48$ integer comparisons; the infinite group never enters. Thresholded at the noise scale this becomes the congruence test of Section 2.7, and completeness is what makes its verdict a certificate rather than a heuristic.[S22]

If A and B are *not* congruent, the minimiser of equation (1) need not present B as an obtuse frame — but it nearly does. A candidate achieving value D has frame conorms, read in A's reduced superbase coordinates, that differ from A's non-negative ones by bilinear forms of the residual, each bounded by $\sqrt{3}\,D$; the minimising frame therefore penetrates the obtuse cone by at most $\sqrt{3}\,D$, with an observed worst case of $0.70\,D$ over random reduced pairs.[S33] This single bound collapses the infinite minimisation into the *widened* Selling closure: enumerate B's superbases with conorms admitted down to $-\sqrt{3}\,D_0$, where D_0 is the plain-closure minimum; the value can only drop, so the widening needed only shrinks, and the search stops with a certificate the moment the widening used covers the value found. One case closes in a single step: when D_0 is smaller than every strictly positive conorm of both lattices, no frame within that distance of either can cross a face, and D_0 is already the ideal.[S31]

A second, independent certificate bounds the search from the eigenvalue side, and it is the one that connects to tradition: the familiar list of 6 960 small matrices — the unimodular matrices with entries in $\{-1, 0, 1\}$, 6 960 of the $3^9 = 19\,683$ candidates; allowing entries up to ± 2 gives 135 408.[S33] If a candidate achieves value D in

equation (1), then comparing the two quadratic forms on every vector gives

$$\|\mathbf{M}\|_2^2 \;\le\; \frac{\lambda_{\max}(\mathbf{G}_A) + D}{\lambda_{\min}(\mathbf{G}_B)} \;=:\; R(D)^2, \tag{2}$$

where $\lambda_{\max}(\mathbf{G}_A)$ and $\lambda_{\min}(\mathbf{G}_B)$ are the largest and smallest eigenvalues of the two metric tensors and $\|\mathbf{M}\|_2$ is the spectral norm of $\mathbf{M}$; every entry of $\mathbf{M}$ is bounded by $\|\mathbf{M}\|_2$, so any matrix that could improve on D lies inside the entry ball of radius $R(D)$. A single pass closes the search: the running minimum only shrinks, R shrinks with it, and the ball enumerated at the seed's radius therefore already contains every possible improver. The 6 960 list is the radius-one instance of this data-sized ball, sufficient exactly when $R(D) < 2$, that is, when $\lambda_{\max}(\mathbf{G}_A) + D < 4\,\lambda_{\min}(\mathbf{G}_B)$ — a per-pair inequality the data verifies, not a definition. For decades that sufficiency was an assumption; here it is a number the pair itself certifies or overrules — and on the Scotty benchmark of McCoy *et al.* (2026), seven of the seventeen cells have $\lambda_{\max} \ge 4\lambda_{\min}$, so the radius-one list runs uncertified on them at every error level (Section 3.4).[S34] Between two reduced settings of *one* lattice the list always contains a relating operator — a machine-checked proposition over the five stratum-fixed flip graphs, covering every lattice (Appendix D) — but it provably cannot return the whole ambiguity coset: most reduced bases of a V4 or V5 lattice carry self-coset operators with an entry of two, and cuboid settings drawn from different arrangement classes are related by operators with an entry of three, which the closure route produces and the truncated ball cannot (Section 4.3).[S22] Between *different* lattices the certificate is the only guarantee, and anisotropy overrules it: on a reduced needle pair with $R \approx 7$ the radius-two ball improves the radius-one minimum almost threefold.[S22] It is certified on the Hfq pair of deposited cells, where $R = 1.86$, the radius-one ball provably suffices, and the two directed values, 492 and 533 Å^2, are global minima — their difference is the honest asymmetry of anchoring, which the symmetrised report absorbs.[S33] The certificate has one limit: R grows with anisotropy, and for a cell as elongated as the $P6_3$

lattice of Section 4.2, $R \approx 7.9$ already at $D = 0$, the entry ball is infeasible; there the widened closure above is the enumerator — the same ideal, two interchangeable proofs of termination, each cheap where the other is expensive.[S33] The strain tensor of the minimiser, $\varepsilon = \frac{1}{2}\,\mathrm{sym}(\mathbf{G}_A^{-1}\Delta\mathbf{G})$, then gives the anisotropic resolution $d_{\mathrm{info}} \propto \|\varepsilon\| R_{\mathrm{eff}}$ to which the two lattices carry mutual information without rescaling — the quantity that decides merging — while for cross-crystal phasing the affine strain is refined away and the lattice contributes the operator, the realized correlation being measured by intensity statistics (McCoy *et al.*, 2026).[S33]

What the certificate excludes, and in which directions, deserves plain words, because the box over $\{-\lfloor R\rfloor, \ldots, \lfloor R\rfloor\}$ is the certificate at its crudest. The entries of $\mathbf{M}$ are counts: column j builds A's j-th basis vector by stacking that many copies of B's basis vectors, so a large entry builds a long vector, and no stack may exceed A's longest dimension plus the slack that D buys — that ratio is $R(D)$, and the integer alphabet it leaves is $\{-\lfloor R\rfloor, \ldots, \lfloor R\rfloor\}$. But the constraint is not isotropic. Comparing the two quadratic forms entry by entry gives, for each column, $|\,\mathbf{M}_{\cdot j}^{\mathsf{T}}\mathbf{G}_B\mathbf{M}_{\cdot j} - (\mathbf{G}_A)_{jj}| \le D$, and the same for every pairwise dot product: between two near-congruent cells, each column of $\mathbf{M}$ must be a lattice vector of B whose length matches the corresponding cell edge of A to within D, with mutual angles matching the same way. The declared error therefore decides how far to search *per column and per direction* — three thin ellipsoidal shells in B's lattice, not a cube of digits. On the $P6_3$ needle against itself at $D = 50\,\text{Å}^2$ the difference is the whole story: the shells hold 6, 6 and 2 candidate vectors, 72 column triples before the determinant and angle filters, while the entry box at $\lfloor R\rfloor = 7$ holds $15^9 \approx 4 \times 10^{10}$ matrices; and widening to $D \approx 125\,\text{Å}^2$ admits genuinely new long-shell candidates, the in-plane stacks of length 232.7 Å that shadow the 233 Å axis — the pseudo-sublattice ambiguity surfacing exactly where the per-column prescription says it must.[S34] The widened closure above is the final sharpening

of the same prescription: it prices in the angles from the start by keeping only jointly near-obtuse frames, at most 32 members plus flips, independent of anisotropy.

2.3. From the ideal to the key, one relaxation at a time

Equation (1) is exact but pairwise, and an archive needs cheaper objects. Each object below is obtained from the ideal by a single, named modification, with the price of the modification stated; Table 5 summarises the sequence.

Restrict the candidates to nearly obtuse frames. By the $\sqrt{3}\,D$ bound of Section 2.2, discarding frames that penetrate the obtuse cone deeper than $\sqrt{3}$ times the current minimum loses nothing. The candidate set shrinks from the integer ball to the widened Selling closure, the minimum decreases towards the ideal as the admitted penetration grows, and the search stops with a certificate the moment the admitted penetration exceeds the current minimum. This is the rung that remains feasible where anisotropy defeats the entry ball of Section 2.2.

Restrict further, to penetration zero. Jointly obtuse frames only: the Selling-closure minimum, at most 32×32 cached candidates. The price becomes visible rather than hidden: the restriction is exact whenever the minimum is smaller than every positive conorm of both lattices, since no frame within that distance of either can cross a face, and it is an upper bound otherwise, discontinuous across faces — Appendix D exhibits the failure and its magnitude.

Change the question from a value to a decision. Thresholding the ideal at the declared noise scale asks only whether two cells are congruent within measurement error, and the argmin set is then the operator with its automorphism coset. That is the congruence test of Section 2.7, with the guarantee stated there: every tolerance a stated multiple of one noise scale, nothing true lost, nothing false accepted.

Relax the objective to one stored vector per lattice. Retrieval cannot afford a min-

imisation per pair. On the closure, the metric tensor and the conorms are linear re-coordinatisations of one another, and every member of a lattice's closure carries the same six conorms in some order; discarding the order — sorting — turns the pairwise aligned comparison into the difference of two precomputed vectors. That is the key of the next subsection, with its properties inherited from this construction: one vector per lattice by closure invariance, a lower bound on the aligned comparison by the rearrangement inequality, exactness at the ideal's zero set, and a finite many-to-one collapse as the price.

Truncate the ideal's ball at radius one. The classical 6 960-matrix search, valid exactly when the certificate radius of equation (2) is below two — a condition the data verifies per pair. Between reduced settings of one lattice the truncation provably always finds *a* relating operator, yet provably misses parts of the V4 and V5 ambiguity cosets (Appendix D); between different lattices anisotropic pairs overrule it outright (Section 4.3). A Le Page search, finally, approximates the ideal's self-distance coset by angular residuals, and the sequence above subsumes it. The two results of Section 4 are the two ends of this derivation at work: the key powers the atlas, and the thresholded ideal powers reindexing.

2.4. The sorted six-number key

The archive search key divides each conorm by the square root of their sum and sorts the six numbers:

$$\mathbf{s}(\Lambda) = \operatorname{sort}\left(p_{ij}/\sqrt{T}\right), \qquad T = \sum_{i<j} p_{ij}. \tag{3}$$

T is unchanged by a Selling move at a zero conorm, so every member of the closure carries the same six numbers in some order and every basis of a lattice yields the same key.[S5] The map is linear in the conorms, so the key is Lipschitz in the metric tensor, and it scales with length: $\mathbf{s}(\lambda\Lambda) = \lambda\,\mathbf{s}(\Lambda)$. A cube of edge a has key

$(0, 0, 0, a/\sqrt{3}, a/\sqrt{3}, a/\sqrt{3})$, and stretching every edge by Δa moves the key by exactly Δa. Sorting discards which conorm belonged to which pair, so the distance is the rearrangement lower bound of Lemma 1: the sorted distance never exceeds the aligned distance of any superbase pairing.[S7] For shape-only comparison the key is divided by $V^{1/3}$.

Kurlin's root products replace $p_{ij}/\sqrt{T}$ by $\sqrt{p_{ij}}$. They remain available on request and are not the archive key. At a vanishing conorm the square root is only Hölder-$\frac{1}{2}$. On the Scotty test cells of McCoy *et al.* (2026) the square-root distance is 16.7 and 16.4 Å for 1fe5→1u4j and 1fe5→1g2x, against linear distances 2.7 and 2.9 Å, and 13.8 Å against 5.4 Å for 4nl7→2yht. NCDist ranks these three pairs as the linear key does — the two 1fe5 pairs close together and the Hfq pair clearly farther — while the square-root key reverses that order; NCDist values carry an arbitrary internal scale factor, so they are compared with other metrics only by ranking, never by magnitude.[S29]

The key is Euclidean in a substantive sense. Its distance is the quotient distance of $\mathbb{R}^6$ by *all* permutations of the six entries (Lemma 1), and because that permutation group is generated by reflections the quotient is the convex cone $\{s_1 \leq \cdots \leq s_6\}$ with the ordinary flat metric.[S8] The physically exact relabelling group — the image of S_4 in S_6 — is smaller and contains no reflections, so its quotient, the space in which Kurlin's complete invariant lives, is not Euclidean; this is why a metric has to be constructed there separately (Appendix B). Because the sorted key lives on a flat cone, KD trees, neighbour graphs and principal components apply to it without modification.

What the key is not is a fingerprint. With the pairing gone, a few distinct lattices can share a key: generically 30 for triclinic (V1) lattices, 15 for V2, 4 for V4, and none for V3 and V5 (Appendix B).[S9] This is harmless provided equality of keys is never used as a proof of identity. The key is a filter, and it is a safe one because of the

following statement, proved in Appendix B.

Lemma 1 (conservative sorted-key distance) *Let $x, y \in \mathbb{R}^6$ and let $G \subseteq S_6$ be any set of permutations. Then*

$$\| \operatorname{sort}(x) - \operatorname{sort}(y) \|_2 = \min_{\sigma \in S_6} \|x - \sigma y\|_2 \leq \min_{\sigma \in G} \|x - \sigma y\|_2. \tag{4}$$

With G the image of S_4, the key distance is a lower bound on the orbit distance between the obtuse presentations of two lattices — the aligned distance over their Selling closures.[S7] That is the distance the retrieval guarantee refers to throughout; the deformation-tolerant distances of Appendix D, which admit frames outside the obtuse cone, are not bounded by the key.[S31] A radius query on keys therefore returns every lattice within that radius of the query, plus occasionally a few impostors, which the next stage removes. A fixed-k nearest-neighbour query carries no such guarantee, because key collisions can crowd the list; it remains a useful exploratory tool.

2.5. A representative can jump while the lattice does not

Along the synthetic monoclinic series ($a = 120$, $b = 189.1$ Å, $\beta = 91.2°$, c from 150 to 100 Å) the metric changes smoothly. The Selling closure stays at 12 obtuse superbases, the V4 count, through $a = c$ (Fig. 1). In the equal-perturbation test about $a = b$, reference $(40, 40, 60, 90, 91, 90)$ against neighbours shifted by ± 0.001 Å in b, the raw reduced distance is 0.080 on one side of the boundary and 118.5 on the other, while the sorted-key distance to the same reference is 0.0010 on both sides.[S6] An earlier raw-distance figure of 83.7 is not reproduced and is not used. The jump is the arrangement ambiguity that Andrews *et al.* (2019) themselves acknowledge: a single S^6 representative is one of 24 (or more) arrangements, so raw S^6 distance is not a lattice distance, and their companion treatment restores one by minimizing over reflections and boundary transforms. The sorted key removes that minimization by construction,

at the cost of uniqueness. The discontinuity belongs to the representative, not to the lattice, and the point $a = c$, where an axis exchange becomes an exact symmetry, is a junction that the search coordinate passes through smoothly while the exact stage exposes the extra operators that become admissible there.

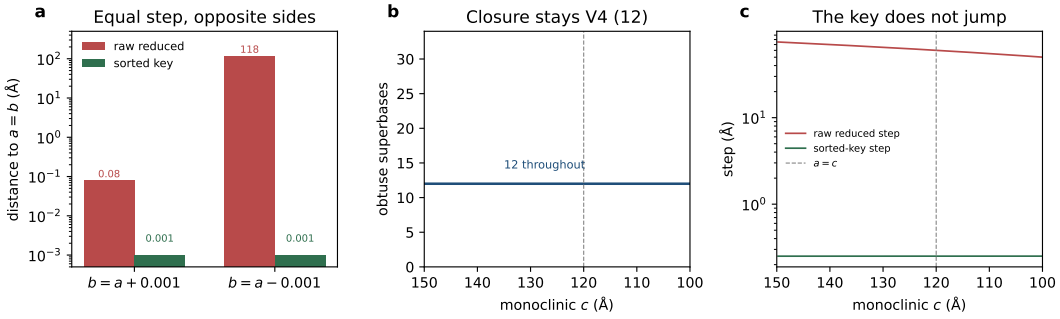


Fig. 1. A representative can jump while the sorted key does not. (a) Equal perturbation of 0.001 Å on either side of $a = b$: raw reduced distance 0.080 versus 118.5, sorted-key distance 0.0010 on both sides. (b) Along the monoclinic series the Selling closure stays at 12 superbases through $a = c$. (c) Consecutive raw reduced steps stay tens of angstroms; consecutive sorted-key steps stay near 0.25 Å.

2.6. Pseudo-symmetry near right angles

The square-root products of Section 2.4 are Hölder-$\frac{1}{2}$ where a conorm vanishes — at a right angle in the obtuse superbase, which every orthorhombic, tetragonal, hexagonal and monoclinic P lattice has exactly, and every pseudo-symmetric cell has nearly. The consequence is deterministic before it is statistical: two distinct lattices that both lie near such a right angle are close in the metric tensor, but their square-root keys separate as the square root of that closeness, so the square-root distance inflates exactly the neighbourhoods where the archive is densest. On monoclinic P cells $(50, 70, 60, 90, \beta, 90)$ the pairs $\beta = (90{+}\delta,\ 90{+}2\delta)$ are 0.80, 0.080 and 0.008 Å apart in the key of equation (3) for $\delta = 1°$, $0.1°$ and $0.01°$, but 3.1, 0.95 and 0.30 Å apart in the square-root key — an inflation that grows without bound as $\beta \to 90°$.[S29] The

Scotty cells show the same effect between deposited lattices (Section 2.4; Appendix C). Measurement noise is the stochastic form of the same cusp: on the CXIDB 83 hexagonal cell a 0.01° angular error moves the square-root key by about 1.4–1.6 Å, ten times the c-axis spread measured in Section 3.2.[S11] The archive key of equation (3) has neither problem: it is Lipschitz in the metric tensor. Floored and soft-threshold maps remain noise models for serial frame data; they are not used for archive search (Appendix C).[S12]

2.7. Exact certification and reindexing over the closure

The usual way to reindex a new cell N onto a reference R is to find the metric symmetry of the lattice by a Le Page-type search (Le Page, 1982): enumerate directions that are twofold axes of the metric to within an angular tolerance, build the point group, and read admissible reindexing operators off it. That procedure answers a different question from the one reindexing asks. It finds operators that are near-symmetries of one metric, which is what pseudo-symmetry and twin-law analysis need; reindexing needs the operator that relates two settings, and the two coincide only when the lattice is exactly symmetric. Away from exact symmetry the relating operator is still an exact integer matrix, but it drops out of the tolerance-enlarged group as the lattice moves away from the symmetric locus (Section 3.3),[S20] and widening the tolerance admits unrelated pseudo-symmetries instead. Reduction flips make matters worse: a reduced-cell algorithm can hand back R and N in different settings for reasons that have nothing to do with symmetry.

The procedure used here never asks how symmetric the lattice is. It asks only whether an integer basis change maps one metric onto the other, over a finite list of candidates that is complete by construction. Its steps are as follows; Appendix D gives the pseudo-code and the form of the operator.

1. *Go primitive, and remember how.* If R or N is given in a centred conventional setting, convert to a primitive basis with the rational centering matrix $\mathbf{C}$, $\mathbf{B}_{\mathrm{prim}} = \mathbf{B}_{\mathrm{conv}}\mathbf{C}$, and keep $\mathbf{C}$. This is the only rational step; everything after it is integer.

2. *Selling-reduce, recording integer coordinates.* Reduce each primitive basis to an obtuse superbase and keep the four superbase vectors as integer triples in the primitive basis ($\mathbf{U}_R$, $\mathbf{U}_N$).

3. *Enumerate the closure.* From the zero pattern of the conorms, with zeros detected at a tolerance derived from the noise level (Appendix C), enumerate $\mathcal{S}(R)$ and $\mathcal{S}(N)$. For the reference this is done once and cached; it contains at most 32 members.

4. *Optional key screen.* If the sorted keys differ by more than the noise-derived radius, N is not a setting of R and no operator search is needed.

5. *Form every candidate operator.* For each pair of closure members, each of the 24 relabellings and each overall sign, take three of the four superbase vectors as columns of integer matrices $\mathbf{U}'_R$, $\mathbf{U}'_N$ and set $\mathbf{P} = \mathbf{U}'_R\,\mathbf{U}'^{-1}_N$. Every $\mathbf{P}$ is an integer matrix of determinant ± 1.

6. *Keep those that carry the metric.* Accept $\mathbf{P}$ if $\|\mathbf{P}^{\mathsf{T}}\mathbf{G}(R)\mathbf{P} - \mathbf{G}(N)\|$ is below a residual tolerance calibrated on the same noise model as step 3. The residual is linear in the metric.

7. *Report the coset, not one matrix.* The accepted set is $\mathbf{P}_0 H$ with H the automorphism group of the lattice. Return all of it, separated by determinant sign if handedness matters. Choosing among the members is not a geometric question; intensity statistics decide, or the ambiguity is carried forward. When the space

group is known, operators differing by an element of the Laue group L give the same intensities, so the distinct indexing alternatives are the cosets H/L, $|H|/|L|$ of them, and one representative per coset is passed to the intensity test (Lebedev *et al.*, 2006; Brehm & Diederichs, 2014; Gildea & Winter, 2018).

8. *Propagate.* With $\mathbf{B}_N = \mathbf{B}_R\mathbf{P}$: fractional coordinates $x_R = \mathbf{P}\,x_N$, Miller indices $h_R = \mathbf{P}^{-\mathsf{T}}h_N$, symmetry operators $W_R = \mathbf{P}W_N\mathbf{P}^{-1}$; in conventional settings $\mathbf{P}_{\mathrm{conv}} = \mathbf{C}_R\,\mathbf{P}\,\mathbf{C}_N^{-1}$, rational with the centering denominators.

Nothing in steps 1–8 is rounded to an integer; the operators are integer because they are built from integer coordinates, and floating point enters only in the comparison of step 6. Symmetry tolerance is not used to find the operator. The one tolerance that remains decides whether two metrics are the same lattice and, at a symmetric locus, how large the coset is (Appendix D).

Everything that depends on the reference is computed once. For a fixed R the centering matrix, primitive metric, reduced superbase, the complete closure with integer coordinates, the sorted key and its radius, and the automorphism coset $H = \mathrm{Aut}(R)$ obtained by running steps 5–6 of R against itself are cached. The full closure has at most 32 superbases, so the cache is small, and it is exactly the object that the "24 relabellings" shortcut omits. A new frame N then costs one Selling reduction, one closure enumeration (needed only if N is itself near a symmetry stratum; otherwise its closure is the single reduced superbase), and at most $32 \times |\mathcal{S}(N)| \times 48$ integer 3×3 products, each followed by one metric residual. Because H is known in advance, the coset for a frame is reported as $\mathbf{P}_0$ times the known H, and if the space group is known, as $\mathbf{P}_0$ together with $|H|/|L|$ branches — two for $P6_3$, one for $P4_32_12$ and $P2_12_12_1$ — which is what an indexing-ambiguity resolver downstream requires.[S21]

Two statements delimit what this stage does and does not promise. First, the closure match is a congruence test, not a metric: its inputs carry the hypothesis that the

two cells describe the same lattice, possibly measured with noise, and its residual is meaningful under that hypothesis. Read as a distance between genuinely different lattices it misbehaves — Appendix D shows it discontinuous across Selling faces — and the continuous object for similarity is the key.[S31] Second, under the hypothesis the procedure is guaranteed, with every tolerance tied to one declared noise scale. Write η for the conorm-scale noise (σ_θ on the invariant scale of Appendix C) and set the retrieval radius, the zero tolerance and the verification residual to small stated multiples of η. Then the measured pair is retrieved, because the linear key moves by at most order η; the widened closure of each measured cell contains a counterpart of every member of the true lattice's closure, because reduction of a measured cell lands on a presentation of the truth that is obtuse to within η and every flip needed to reach the remaining members triggers; the full automorphism coset is generated and accepted, because each operator has bounded integer entries and therefore residual of order η; and nothing false is accepted, because every surviving operator maps one measured metric onto the other within the stated residual.[S32] What no procedure can decide is whether two cells are congruent within η or distinct by less than it: that boundary is the noise model itself, and the decision belongs to intensity statistics (Lebedev *et al.*, 2006).

2.8. Pseudo-symmetry as cosets between closures

The closure yields the exact automorphism group of R. Operators that are automorphisms of a nearby, more symmetric lattice — the source of pseudo-merohedral twinning and of indexing ambiguity near a symmetry stratum — are not in it, and adding them requires deciding what "nearby" means. Table 2 separates the four sources of ambiguity in assigning indices and where each is resolved. The reduction flip of Section 3.3 is setting ambiguity; treating it as pseudo-symmetry is what makes a Le Page

gate lose it. At exactly $a = c$ the first two rows of the table merge, because the flip operator becomes an automorphism.

For pseudo-symmetry the nearby equality is imposed and the closure used twice. The lattice R^* on the stratum is built, its closure $\mathcal{S}(R^*)$ enumerated and self-matched to give $H^* = \mathrm{Aut}(R^*)$. Then $H \subseteq H^*$, the pseudo-symmetry operators are the non-trivial cosets H^*/H, and the branches that intensity statistics must examine are H^*/L. Each extra operator carries its residual $\|\mathbf{P}^{\mathsf{T}}\mathbf{G}(R)\mathbf{P} - \mathbf{G}(R)\|$, which grows with the distance to the stratum.[S17] The single tolerance is that declared distance, reported alongside the operators. The imposition must act on whole $\mathrm{Aut}(R)$-orbits of coform entries, otherwise R^* depends on which member of the orbit was edited and $H \subseteq H^*$ can fail; Appendix D records the rule. The reported result is the two hand-imposed strata: $a = c$ in a monoclinic family, and the two short edges of a near-tetragonal cuboid. Appendix D records the general pattern loop. The checks do not run that loop over every degeneration pattern; they impose these two strata by hand.

2.9. Data sets and implementation

Archive-scale results use the unit cells of 206 214 PDB entries. The lysozyme family comprises all hen egg-white lysozyme entries (UniProt P00698; 1372 entries, 1361 with a cell). Per-crystal serial data are the CrystFEL stream for a β-lactamase XFEL data set (CXIDB entry 83; 14 445 patterns, 12 474 indexed crystals, space group $P6_3$, cell near $41.8 \times 41.8 \times 233$ Å). Synthetic families are a monoclinic series ($a = 120$, $b = 189.1$ Å, $\beta = 91.2°$, c from 150 to 100 Å) that passes through the pseudo-symmetric region around $a = c$, and 200 perturbed monoclinic frames straddling a reduction flip. All methods are implemented in the `agentsg` Python package, with the KD tree taken from SciPy (`scipy.spatial.cKDTree`); function names in the Supporting Information refer to it, and the pseudo-code of Appendix D is sufficient to re-implement each step

independently.

3. Results

3.1. The PDB in lattice-shape space

Keys were computed for the 206 214 PDB cells and indexed with a KD tree.[S23] On the precomputed keys the index builds in 0.044 s and a median $k = 10$ query takes 0.010 ms (Python 3.12, NumPy 2.5.3, SciPy 1.18.1). Earlier figures of 0.28 s and 0.04 ms are not this measurement. A radius query on that index carries the guarantee of Lemma 1; certification is applied only to the candidates it returns. For scale, the SAUC search of alternate unit cells reported 500-nearest-neighbour queries over roughly half a million PDB and CSD cells in about 4 s with a Selling embedding versus about 8 s with Niggli (McGill *et al.*, 2014); those searches operate in an orbit-aware space, whereas a query here searches the lossy key and relies on the certification stage that follows. There is one key per lattice at every Voronoi type;[S5] the extra superbases of symmetric lattices are never inserted into the index and are opened only for certification.

For shape comparison the key is divided by $V^{1/3}$, a neighbourhood graph is built over the whole PDB, and a two-dimensional embedding is drawn. It is an embedding of the key distance, not of Kurlin's complete lattice-isometry space, and nearby points are nearby by construction. The scientific content is in how the observed population occupies the space: when the embedding is coloured afterwards by crystal system, tetragonal populations form branches that approach cubic symmetry at a restricted locus, orthorhombic populations occupy neighbouring and connecting regions, trigonal and hexagonal regions appear at other boundaries, and monoclinic cells spread over a broader area.[S24] None of these labels were used to build the key. Locally, a mean-centred singular-value decomposition of the volume-normalised keys in $k = 40$

neighbourhoods, on a 30 000-cell subsample, gives an entropy effective rank from 1.00 to 5.69, median 2.02.[S26] A previously printed median of 4.12 is not reproduced by this estimator and is not used. The embedding in Fig. 2 is where the population is displayed. The fraction of key-space edges whose distance lies within 10% of the exact orbit distance has not been reported.[S25] Branches and junctions are not read as the global topology of lattice space until that audit is done.

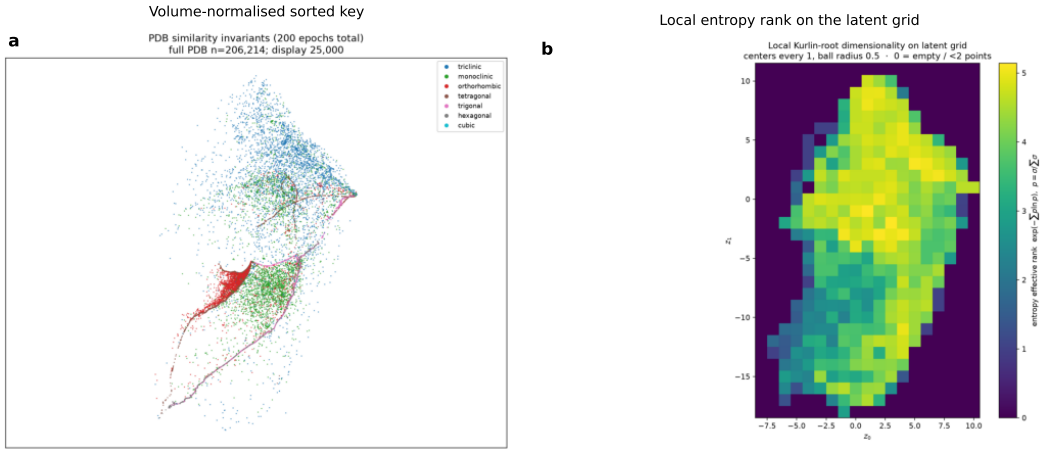


Fig. 2. Archive shape comparison. (a) Two-dimensional embedding of the volume-normalised sorted key for 206 214 PDB cells, coloured afterwards by crystal system; 25 000 cells are drawn. (b) Local entropy effective rank on a unit grid of that latent plane (ball radius 0.5). The neighbourhood estimator cited in the text is a separate $k = 40$ calculation in the six-dimensional key, median 2.02, and is not read off panel (b).

The structure the colouring reveals has an exact, computable skeleton. In the volume-normalised key space each Bravais family is a definite geometric object: the three cubic lattices are single points — cP at $(0, 0, 0, 0.577, 0.577, 0.577)$, cI at $(0.257, \ldots, 0.257)$ and cF at $(0, 0, 0.397, 0.397, 0.397, 0.397)$ — the tetragonal, rhombohedral and hexagonal families are one-parameter curves, the orthorhombic families two-parameter surfaces, and so on down to triclinic, which fills the five-dimensional shape cone.[S30] The curves meet the points at exact nexus: cP is a quintuple point — the terminus of the

tP curve, the corner of the oP surface, the end of the oC family through its tP edge, and the $\alpha = 90°$ point of the rhombohedral curve; hR meets tI at exactly two points, cI at ($\alpha = 109.47°$, $c/a = 1$) and cF at ($\alpha = 60°$, $c/a = \sqrt{2}$), and the two nexus are fed by different centrings, oI running into the first through oI(a=b) = tI and oF into the second through oF(a=b) = tI at axial ratio $\sqrt{2}\,c/a$.[S30] Distinct families also run together without meeting, along asymptotic seams at extreme axial ratio: hP at $c/a = R$ runs alongside hR at hexagonal-setting $c/a = 3R$ (ABC stacking at spacing $c/3$ against AA stacking at c), with key distance 0.093, 0.014 and 0.0009 at $R = 1$, 4 and 32; and tI at $c/a = 2R$ runs alongside tP at $c/a = R$ (a body-centred cell is two interleaved P lattices at half the repeat), with distance 0.64, 0.15 and 0.004 at $R = 1$, 4 and 64, plus a mirror seam at the flat end through the 45° relation tI($R/\sqrt{2}$).[S30] The two tetragonal curves are farthest apart near the cubic points, 0.50 from cP to the tI curve and 0.71 from cI to the tP curve, which is why the embedding shows the tetragonal branches splitting at the cubic pinch and re-approaching at both extremes.[S30] Table 4 collects the skeleton. A population bridge at a nexus means lattices can pass through the symmetry; a bridge along a seam means two families merely become metrically indistinguishable at extreme aspect. The embedding renders both as branches approaching; the skeleton tells them apart.

3.2. The atlas at family and per-crystal scale

The same coordinate zooms from the archive to one protein family and to the individual crystals of one experiment.

The 1361 hen egg-white lysozyme cells span six space groups: 1152 tetragonal $P4_32_12$, 100 orthorhombic $P2_12_12_1$, 59 monoclinic $P2_1$, 41 triclinic $P1$, and small trigonal and hexagonal groups.[S27] The key handles all of them in one setting-independent space before any system-specific analysis is chosen. Within the 1113 native-volume

tetragonal cells, which span a 26% range in volume, the leading intrinsic coordinate of the key-distance graph (Fiedler, 1973) correlates with cell shrinkage at $|r| = 0.92$: a varies smoothly from about 76.8 to 80.0 Å while c stays flat and then steps between preferred values near 37 and 38 Å.[S27] For this single-setting subset the same trend could be read directly from (a, c); the gain is that the subset was isolated from a heterogeneous census by one lattice-level distance, after which conventional interpretation proceeds in whatever coordinates suit it.

Merged depositions keep only a consensus cell; the CrystFEL stream retains every indexed cell. Because these are unconstrained per-frame cells of a high-symmetry lattice, they are keyed by sorted conorms without the square root (Section 2.6) and the cloud is analysed by principal components rather than a nonlinear embedding. The c axis has median 233.3 Å, median absolute deviation 0.13 Å, and a maximum of 245 Å. The key is blind to orientation, so orientation metadata can be projected onto the variation as an independent diagnostic (Fig. 3). Crystals with the beam within 10° of c^* ($n = 141$) have a c-axis median absolute deviation of 0.22 Å, against 0.083 Å for beam angles above 80° ($n = 2866$), a ratio of 2.6.[S28] The earlier scatters of 1.53 Å and about 0.5 Å are not reproduced under this binning and are not used. The same projection could be made for pulse energy, detector geometry, time delay or temperature without touching the lattice representation.

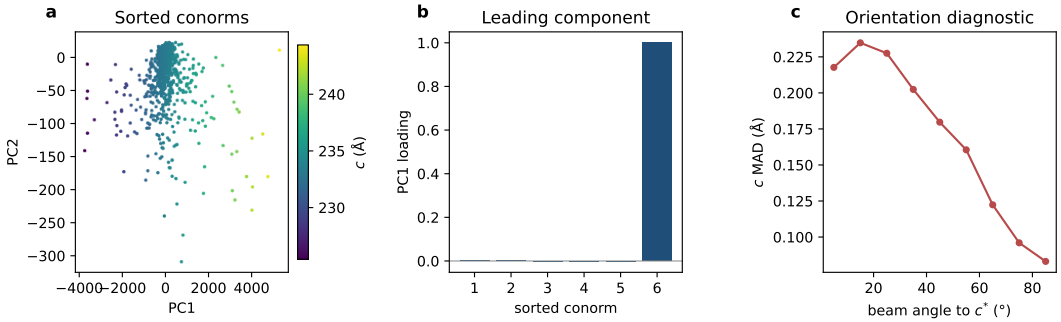


Fig. 3. Per-crystal cells from CXIDB 83, keyed by sorted conorms. (a) Principal-component scores of 4 000 crystals, coloured by c. (b) Loadings of the leading component, carried by the largest sorted conorm. (c) Median absolute deviation of c against the angle between the beam and c^*.

3.3. Exact reindexing over the Selling closure

A conventional reindexing search asks which small integer matrix maps one cell onto another, mixing "is this the same lattice?" with "which basis?". Here they are separated by the procedure of Section 2.7. On an orthorhombic P cell presented in an even-class basis $\{v_1, v_3, v_2 - v_1\}$ the procedure returns the full eight-member *mmm* coset, whereas the 24 relabellings of a single reduced superbase return nothing;[S14] on an arbitrary unimodular resetting of a reference cell, with and without noise and centering, it recovers the resetting up to automorphism for eight lattice types.[S15]

On the 200 perturbed monoclinic frames spanning both sides of a reduction flip, a brute-force baseline tested all 6960 unimodular matrices with entries in $\{-1, 0, 1\}$ per frame. The key-first workflow reduced each frame to a cached two-operator coset, with the reference closure enumerated once before the first frame.[S21] A rerun of that appendix procedure does not recover the earlier wall-clock figures: on this machine the 6960-matrix baseline has median 15 ms per frame and the cached procedure median 75 ms per frame. The figures 0.07 ms per frame and 330-fold are not reproduced and

are not used. The baseline is a cost comparison, not a correctness reference. A search over matrices with entries in $\{-1, 0, 1\}$ finds the relating operator only if that operator has no entry of magnitude two or more; for arbitrary input settings this fails outright, since $\mathrm{GL}(3, \mathbb{Z})$ is infinite and most random resettings with entries up to ± 2 are invisible to it (192 of 200 in the test).[S22] The usual defence is to reduce both cells first. With Niggli reduction the defence works empirically: for two Niggli cells of the same lattice, exact or noisy, in random input settings, every relating operator had entries in $\{-1, 0, 1\}$ across eight lattice types and under adversarial anisotropic and near-degenerate resettings,[S22] consistent with the classical fact that Buerger-cell boundary transformations and point-group operations in a reduced basis are all small. With Selling reduction the two halves of the question separate, and both halves are provable. Finding *one* operator always succeeds: between any two Selling-reduced settings of any lattice some relating operator has entries in $\{-1, 0, 1\}$, by enumeration of the five stratum-fixed flip graphs (Appendix D).[S22] Returning the *coset* provably fails: the ambiguity coset of a single Selling-reduced V4 or V5 cell against itself already contains operators with an entry of two (18 of 24 and 60 of 64 reduced bases, respectively), and two cuboid settings from different arrangement classes — cells $(7.07, 4, 5, 90, 135, 124.4)$ and $(6.40, 4, 5.83, 90, 132.0, 128.7)$, both reduced presentations of the $3 \times 4 \times 5$ cuboid — are related by an eight-operator coset of which the small search returns four and silently drops the rest, including an entry-three operator.[S22] The brute force is therefore complete only conditionally — on reduction for the single operator, and for the coset not at all in V4 and V5 — whereas the closure route is coset-complete unconditionally, by enumeration, and needs at most 32 candidates rather than 6960.[S21]

Near a pseudo-symmetric boundary a tolerance-gated symmetry search can drop a basis change that is needed only because reduction picked a different representative.

In the controlled monoclinic family, exchanging a and c is an exact integer relation between two settings of the same lattice for every $|a-c|$, but an exact metric symmetry only at $a = c$. With a 3° Le Page tolerance the exchange is admitted up to about $|a - c| = 2.85\,\text{Å}$ and then excluded, although the two descriptions remain exactly related; raising the tolerance merely admits unrelated pseudo-symmetries. The closure-based operator search recovers the exact exchange regardless of how far the lattice is from symmetric (Table 3).[S20] The same family, with $a = c$ imposed, gives H^* of index 2 over H and recovers the exchange with residual proportional to $|a - c|$ at every distortion; the near-tetragonal cuboid with its two short edges equalized gives index 2 and a four-fold outside H.[S17]

Matching only one representative per isometry class returns one operator per class pair and loses the coset. On exact cells, an orthorhombic P cell against itself returns the order-8 mmm coset when matched over the full 32-member closure and only 2 operators when matched over representatives; the $P6_3$ cell against itself returns 24 ($6/mmm$ including $-I$) versus 4.[S14] Too tight a verification tolerance lets noise select a subset of the coset: on $P6_3$ frames with 0.05° angular noise, a relative tolerance of 10^{-4} returned 12 of the 24 operators and 10^{-3} returned all 24.[S18] With exact zero detection a measured $P6_3$ frame is classified as V1 and only the 24 relabellings of one superbase are matched; widening the zero tolerance to a few σ on the invariant scale of Appendix C restores the V4 classification and the full closure, and two independently perturbed frames at 0.01° and 0.05° angular noise then both recovered all 24 operators.[S19]

3.4. Relating reduced cells under a declared error

A controlled comparison of the two reduction routes starts from one cell and one declared error. Take a Scotty base cell, reduce it, and perturb its Niggli metric tensor

by $\Delta\mathbf{G} = D\,E$ with E a random symmetric matrix of unit Frobenius norm: the error radius D is then measured in the ideal's own norm, so D, the widening $\sqrt{3}\,D$ and the certificate radius $R(D)$ of equation (2) are all in Å^2 and directly comparable. Reducing the perturbed lattice and relating it to the original is exactly the task a pipeline faces when two measurements, two programs, or two deposits present the same crystal form.[S34]

Three facts emerge. First, the representative jumps long before the error is interesting: at $D = 20$ Å^2, three parts in a thousand of $\|\mathbf{G}\|$, the Niggli basis of the 2yht cell lands in a different reduction chamber in more than half of random directions (14 of 25) — the parameters stay continuous, but the operator relating the two Niggli frames stops being a signed permutation, so "two related Niggli cells in different settings" is the common case, not the corner case.[S34] Second, both routes recover the relation: the radius-one ball finds a value at most D in every run, and so does the widened closure at $\sqrt{3}\,D$.[S34] Third, the routes differ in who can prove it. The ball's proof is $R(D) < 2$, an anisotropy budget $D < 4\lambda_{\min} - \lambda_{\max}$ that each cell carries: 1 739 Å^2 for 2yht, where the certificate holds comfortably and erodes only as D approaches the budget, but *negative* for seven of the seventeen Scotty cells — both lysozyme cells among them, with $\lambda_{\max}/\lambda_{\min} = 5.5$ and a budget of -1077 Å^2 for 2d4k — whose radius-one searches therefore run uncertified at every error level, right answers included: in fifteen perturbed runs on 2d4k the ball recovered the relation every time and could certify it never, while the widened closure recovered it every time with its certificate intact.[S34] Table 7 gives the budget for every Scotty base. To restore the guarantee the ball must grow to radius $R(D)$, with the candidate count rising as the ninth power of the radius, whereas the widening certificate of Section 2.2 is independent of anisotropy: declare D, widen by $\sqrt{3}\,D$, and the search is complete for every lattice pair at the same enumeration cost.[S34]

4. Discussion

4.1. The problem changed, so the object changed

Reduction methods are sometimes compared as competing answers to one question. It is more useful to see them as answers to different questions. The classical question — which cell should represent this lattice? — is answered well by Niggli reduction, and its seams are the unavoidable price of picking one representative. The modern question — which lattice is this, which lattices are near it, and only then which basis is needed? — is better served by a finite closure for the exact part and a coarse continuous key for the search part. The order of computation is: enter primitive lattice space; build the closure appropriate to the Voronoi type; retrieve with one key per lattice; certify candidates over the full closure; propagate the exact operator to the data that need it. This complements Niggli reduction; it does not replace it.

The difference between the two questions is also the difference between what can be proved. Selling reduction *closes*: the set of reduced frames is finite, enumerable, and the same integer object for every lattice of a Voronoi type, so statements about reduced settings — the entry bounds of Appendix D among them — reduce to one finite check per type. Niggli reduction *selects*: it picks a single extremal representative by inequalities among lengths, and there is no closure equivalent — the nearest analogue, the set of Buerger cells, changes its membership and count as the parameters move, chopping each stratum into the case structure that Gruber (1973) mapped by hand and Andrews & Bernstein (2014) mapped by Monte Carlo search, with completeness in the latter argued empirically rather than proved. Everything hard about Niggli reduction — the boundary instability, the many-case boundary polytopes, the entry bound for its relating operators that remains folklore — is the cost of selecting one representative instead of keeping the closed set. The guarantees of the closure route are not cleverness; they are what a set-valued reduction yields by construction.

4.2. A lower-bound key is enough

Lattice-isometry space has not been equipped with a Euclidean metric here; Kurlin (2026) explicitly postpones that construction. What is available is weaker and sufficient: search on an object whose distance lower-bounds the true one, refined by an exact procedure afterwards — which is exactly what a certified filter needs. The exploratory tools that run on the key — nearest neighbours, principal components, embeddings — are descriptive, and their validation is done upstream by auditing key collisions and graph edges against exact comparisons. When more than a bound is wanted, the ladder tops out at the certified ideal of Section 2.2.

4.3. Which tolerances remain, and what they decide

The Le Page-style bet — that an exact integer relation between two settings is found only if it lies inside a symmetry group enlarged by an angular tolerance — is gone, because the closure search is finite and complete. Two tolerances remain, of different kinds. The closure tolerance decides which conorms count as zero for enumeration; being generous costs at most a few extra candidates, each of which is verified, so it is set well above the noise and is not a bet. The verification tolerance decides whether two metrics are the same lattice and, on a symmetry stratum, which automorphisms count. This is the indexing ambiguity itself, which no geometry resolves; it is calibrated on a linear residual from known same- and different-lattice pairs, and the safe direction is to accept the whole coset and pass it to intensity statistics. Pseudo-symmetry, when wanted, is a declared distance to a stratum rather than a per-operator angular gate.

4.4. Continuity does not abolish crystal systems

Higher-symmetry lattices sit on lower-dimensional loci of lattice space where extra metric equalities hold and extra automorphisms appear. The PDB population is nat-

urally read as a continuous object stratified by these familiar discrete classes, and a trajectory may approach, cross or leave a stratum. A representation built around one canonical cell makes such transitions look like jumps in description; a lattice-level coordinate lets the physical path and the changing set of exact basis correspondences be handled separately. The skeleton of Table 4 makes the stratification concrete: strata intersect at exact nexus points, and distinct strata can also run together along asymptotic seams without meeting, so a bridge in the population is read against the computed skeleton rather than from the picture alone.

4.5. Scope and attribution

Three objects must stay distinct: Kurlin's structured root form, which is the complete isometry classification; the sorted key, which is continuous and Euclidean but many-to-one; and the exact operator test over the closure, which establishes identity. No claim here requires one object to be all three. The framework addresses equality, similarity and basis equivalence of primitive lattices; supercell and subcell relations change the lattice and need explicit enumeration of finite-index sublattices when they are part of the search problem.

5. Conclusions

A century of reduction practice has rightly emphasized the question of which cell should represent a lattice. The distance between two lattices has an ideal definition — the smallest symmetrised deformation that makes them congruent, certified by a data-sized integer ball — and the machinery of this paper is a ladder of accelerations that never surrenders the guarantees on the way down. Asking first which lattice this is, and which are near it, yields two results. In one continuous Euclidean coordinate — the sorted, normalised Selling scalars, one key per lattice with a conservative

distance — the Protein Data Bank becomes an atlas: a stratified shape space with a computable Bravais skeleton of points, curves, nexus and seams over which the deposited population drapes, readable at archive, family and per-crystal scale. Over the finite Selling closure, reindexing becomes exact: the integer operator and its full automorphism coset are recovered without a symmetry tolerance, where angular gates and entry-bounded searches fail. The key retrieves, the closure decides, and the lower bound is what makes the hand-off safe. In one line: Euclidean key, then finite Selling closure, then exact operator.

Appendix A
Selling reduction and the Selling closure

A.1. Superbase, conorms, vonorms

Given a primitive basis v_1, v_2, v_3, set $v_0 = -(v_1 + v_2 + v_3)$. The four vectors form a superbase. Their six conorms are $p_{ij} = -v_i \cdot v_j$ for $0 \leq i < j \leq 3$. In the S^6 convention of Andrews *et al.* (2019) the Selling scalars are $s_{ij} = -p_{ij}$ and occupy the non-positive orthant; everything below is written in conorms. The seven vonorms (Conway & Sloane, 1992) are the squared lengths of the four vectors and of the three opposite-edge partial sums,

$$v_i^2 = \sum_{j \neq i} p_{ij}, \qquad v_{ij}^2 = (v_i + v_j)^2 = p_{ik} + p_{il} + p_{jk} + p_{jl}, \tag{5}$$

with $\{k, l\} = \{0, 1, 2, 3\} \setminus \{i, j\}$; they satisfy $v_0^2 + v_1^2 + v_2^2 + v_3^2 = v_{01}^2 + v_{02}^2 + v_{03}^2$ and are the D_7 descriptor of Andrews and Bernstein and Kurlin's voform.

A.2. Reduction

A superbase is obtuse if all $p_{ij} \geq 0$. If some $p_{ij} = -\varepsilon < 0$, the Delone–Selling move replaces the superbase by

$$u_i = -v_i, \quad u_j = v_j, \quad u_k = v_i + v_k, \quad u_l = v_i + v_l, \tag{6}$$

which keeps the zero sum, generates the same lattice, permutes six of the seven vonorms and decreases the seventh by 4ε; the sum of the six conorms decreases by ε (Kurlin, 2026, Lemma A.1).[S1] Applying the move to the most negative conorm until none remains terminates in finitely many steps.[S1] This is the algorithm of Andrews & Bernstein (1988, 2014) and Andrews *et al.* (2019); the conventional-to-primitive conversion that precedes it is the only rational step, and all subsequent operations are integer unimodular.

A.3. The closure

Relabelling the four vectors of an obtuse superbase (S_4, 24 elements) and negating all of them always gives obtuse superbases of the same lattice. Kurlin's Lemmas 4.1–4.5 show that these are all of them only for lattices whose Voronoi cell is the generic truncated octahedron (V1). For the four degenerate types there are additional obtuse superbases not related by relabelling, obtained from the reduced one by the move above applied at a conorm that is exactly zero. Table 1 lists the counts; they were confirmed by exhaustive enumeration of all obtuse superbases with bounded integer coordinates for one test lattice of each type.[S2] The Voronoi type is read directly from the zero pattern of the reduced conorms: no zeros is V1; one is V2; two on opposite tetrahedral edges is V3; two sharing a vertex is V4; three is V5, and other three-zero patterns cannot occur.[S3] For a symmetric lattice within a type the classes may coincide (the hexagonal lattice has one class rather than three, because its automorphisms relate them); the enumeration handles this automatically.[S2]

For measured cells, a conorm that is exactly zero in the ideal lattice is a small positive or negative number, so the closure enumeration must treat conorms within a noise tolerance of zero as zero. The tolerance is set from the angular noise model of Appendix C on the closure-invariant scale T defined there.

A.4. Relation to the S^6 orbit and boundary transformations

In the S^6 formulation of Andrews *et al.* (2019) a Selling-reduced lattice is a point in the non-positive orthant, and the relabelling freedom of the superbase acts as a 24-element group of coordinate permutations; the orbit of a reduced point under that group is what a single reduced representative can reach. The Selling closure is a different and generally larger object: the full preimage of the lattice in the orthant. Global negation of the superbase leaves every scalar unchanged, so in S^6 the closure has 1, 2, 3, 6 and 16 distinct points for V1–V5 (half the superbase counts of Table 1), falling into 1, 2, 3, 3 and 4 relabelling orbits.[S2] Orbit and closure coincide only for generic V1. At every degenerate type the closure contains orbits that no relabelling reaches, and the difference is operational rather than cosmetic: matched over one orbit, an orthorhombic P cell presented in an even-class basis returns no operator at all, while matched over the closure it returns the full mmm coset.[S14]

The bridge to the S^6 machinery is the set of boundary transformations that Andrews *et al.* (2019) tabulate at each face $s_i = 0$ of the orthant: those transformations are the Selling moves applied at a zero scalar, which are exactly the moves that generate the additional closure members above.[S2] In that language, the Selling closure is the relabelling orbit together with its images under all boundary transformations applicable at the exact zeros, iterated until closed — applied statically once per lattice rather than dynamically inside a distance computation, carried with the unimodular bookkeeping that scalar points cannot hold, and complete by Kurlin's enumeration (Kurlin, 2026,

Lemmas 4.1–4.5). The two uses are complementary: near a face, a distance algorithm consults the transformations for robustness; at a face, the closure enumerates them for completeness, with the tolerance version $\mathcal{S}_\tau \supseteq \mathcal{S}$ covering measured cells.

Appendix B
The sorted key: invariance, continuity, lower bound, collisions

B.1. One key per lattice

Relabelling permutes the six conorms, so the sorted list is unchanged, and central inversion leaves every conorm unchanged. For the additional closure members at V2–V5, Kurlin's Lemmas 4.2–4.5 give the coform of each alternative superbase explicitly as a transposition or permutation of the conorms of the reduced one; hence the multiset of conorms is the same on every member of the closure, and so is $\mathbf{s}$. This is the modern form of the classical statement that the reduced Selling scalars are unique as a list (Bravais, 1850; Delone, 1932; Andrews *et al.*, 2019). Numerically, the 4, 6, 12 and 32 superbases of the V2–V5 test lattices each gave a single sorted key.[S5]

B.2. Continuity

Sorting is 1-Lipschitz. Kurlin's Lemma 6.6 bounds the change of each root product under a basis perturbation of size δ by $\sqrt{2\ell\delta}$, where ℓ bounds the vector lengths. Across a Voronoi-type boundary the reduction may return a superbase near a different closure member, but all members share the limiting key, so the lattice-level key is continuous through the boundary, with the Hölder-$\frac{1}{2}$ rate of Appendix C at the symmetry strata themselves.[S6]

B.3. Proof of Lemma 1

Expanding $\|x - \sigma y\|_2^2 = \|x\|^2 + \|y\|^2 - 2\langle x, \sigma y\rangle$, the minimum over σ is attained when $\langle x, \sigma y\rangle$ is maximal, which by the rearrangement inequality occurs when both vectors are similarly ordered; that value is $\langle \operatorname{sort} x, \operatorname{sort} y\rangle$. The inequality holds because $G \subseteq S_6$.[S7] The physically realisable relabellings of the six conorms form the image of S_4 in S_6, a subgroup of order 24, so the lemma applies with G that image: the key distance is a lower bound on the orbit distance and the tightest lower bound available from the unpaired multiset alone. The same argument applies verbatim to any monotone per-slot transform of the conorms (Appendix C), provided the transform depends only on the conorm and on lattice invariants.

B.4. Why the key is Euclidean and the orbit metric is not

The lemma identifies the key distance with the quotient metric of $\mathbb{R}^6$ by S_6 acting by coordinate permutations. S_6 is generated by the transpositions $(i\,j)$, each the reflection in the hyperplane $x_i = x_j$; for a finite reflection group the quotient $\mathbb{R}^n/W$ is isometric to a closed Weyl chamber, here the convex cone $\{s_1 \le s_2 \le \cdots \le s_6\}$, with the metric inherited from $\mathbb{R}^n$. Convexity makes chordal and intrinsic distance agree, so the key space is a flat Euclidean domain. The image of S_4 in S_6 contains no reflection: a transposition of two vertices moves two pairs of edges, which is a rotation in $\mathbb{R}^6$. Its quotient is an orbifold with singular strata, not a convex cone, and the orbit metric $\min_{\sigma\in S_4} \|x - \sigma y\|$ is not Euclidean-embeddable, as Schoenberg's criterion confirms numerically.[S8] This is the obstruction that Kurlin's deferred metric on the root-invariant space must address; the sorted key sidesteps it by quotienting by the larger reflection group, at the price of the collisions counted next.

B.5. Collisions

The sorted key forgets the pairing of conorms into the three opposite-edge columns of Kurlin's 2×3 root form. For six distinct values the S_4-orbit of a labelled coform has 24 elements and there are $6! = 720$ arrangements, so $720/24 = 30$ non-isometric lattices share a generic V1 key. The same count gives 15 for V2 ($5!/|D_4|$) and 4 for V4 (which of four values is the distinguished one); V3 and V5 are injective. Concretely, the coforms $\begin{pmatrix} 1 & 2 & 3 \\ 4 & 5 & 6 \end{pmatrix}$ and $\begin{pmatrix} 1 & 2 & 3 \\ 4 & 6 & 5 \end{pmatrix}$ are non-isometric lattices with identical keys.[S9]

B.6. A sharper key

Kurlin's Example 6.5 exhibits non-isometric lattices with identical vonorm multisets (D_7 alone is incomplete) but different conorm multisets, which the sorted key separates. The converse also holds empirically: in 300 random generic V1 fibres, all 30 lattices sharing a sorted key had distinct sorted-$\sqrt{D_7}$ keys.[S10] The concatenated key $\mathrm{sort}(r) \,\|\, \mathrm{sort}(\sqrt{\text{vonorms}}) \in \mathbb{R}^{13}$ is still a pure sort, still continuous, still satisfies Lemma 1 blockwise, and reduces the generic over-retrieval to essentially one. This is an observation rather than a theorem; ties and degenerate types need separate treatment, and the six-dimensional key remains the archival key of this paper.

Appendix C
Noise: the Hölder-$\frac{1}{2}$ cusp and how to remove it

C.1. The effect

Angular measurement error is approximately Gaussian in conorm space, because p_{ij} is linear in the metric tensor: $\sigma_{p_{ij}} \approx |v_i||v_j|\sigma_\theta$.[S11] The square root maps this to the key. Where $p > 0$ the map is locally linear with slope $1/(2\sqrt{p})$; where $p = 0$ it satisfies

only $|\sqrt{x} - \sqrt{y}| \le \sqrt{|x-y|}$, so the response to noise is proportional to $\sqrt{\sigma_p}$ rather than to σ_p. Every lattice with a right angle in its obtuse superbase has such a zero. On the $P6_3$ cell $(41.8, 41.8, 233, 90, 90, 120)$ the numbers are in Table 6: a 0.01° error moves the plain key by a median 1.58 Å, while the same error on a generic triclinic cell moves it by roughly 0.03 Å.[S11] The cusp acts on real separations, not only on noise: for two distinct lattices whose shared near-zero conorm differs by δ, the square-root keys separate as $\sqrt{\delta}$ while the metric tensors, and the linear keys, separate as δ — the monoclinic numbers of Section 2.6.[S29]

C.2. The linear search key

The archive search key is $\mathrm{sort}(p_{ij}/\sqrt{T})$ with $T = \sum_{i<j} p_{ij} = \frac{1}{2}\sum_i |v_i|^2$. A Selling move at a zero conorm permutes the conorms, so T is closure-invariant and the sorted key is one vector per lattice.[S13] The four properties are that invariance, the rearrangement lower bound of Lemma 1, the Lipschitz dependence on the metric, and the length scaling $\mathbf{s}(\lambda\Lambda) = \lambda\,\mathbf{s}(\Lambda)$. A cube of edge a has nonzero slots $a/\sqrt{3}$, and an isotropic edge change Δa moves the key by Δa. The Hölder-$\frac{1}{2}$ cusp of $\sqrt{p}$ is why the square-root products are not the archive key: on the Scotty cells of McCoy *et al.* (2026), 1fe5→1u4j and 1fe5→1g2x are 16.7 and 16.4 Å in the square-root key against 2.7 and 2.9 Å linear, and 4nl7→2yht is 13.8 Å against 5.4 Å. The linear key reproduces NCDist's ranking of the three pairs and the square-root key reverses it; NCDist's values are on an arbitrary internal scale and enter only ordinally.[S29]

C.3. Noise models for serial data

Floored and soft-threshold maps are for frame-level serial data, not for archive search. The noise floor s must be closure-invariant. Per-pair floors $|v_i||v_j|\sigma_\theta$ are not: the even-class superbases of a cuboid have different edge lengths, and keys computed

from two bases of the same cuboid then differ by about 0.02 Å.[S13] With the invariant scale T above, the floor is $s = \sigma_\theta T$. The floored root $\sqrt{p+s} - \sqrt{s}$ equals $\sqrt{p}$ for $p \gg s$ and is linear with slope $1/(2\sqrt{s})$ for $p \ll s$. The soft threshold $\sqrt{\max(p - \kappa s, 0)}$ is a positive-part shrinkage onto the symmetry stratum, deliberately blind to deviations from symmetry below $\kappa\sigma_p$. Table 6 gives their measured behaviour.[S12] Choosing s chooses the resolution below which "symmetric" and "nearly symmetric" are no longer distinguished, and Lemma 1 then holds in that stabilized metric.

Appendix D
Exact certification, cosets and pseudo-code

D.1. Operator construction

During reduction the four superbase vectors are tracked as integer coordinate triples in the input primitive basis, for every member of the closure. For a candidate pair (A, B) let $\mathbf{U}(A)$ and $\mathbf{U}(B)$ be 3×3 integer matrices whose columns are three of the four vectors of a superbase from $\mathcal{S}(A)$ and $\mathcal{S}(B)$ respectively, under one of the 24 relabellings and one sign. Any three vectors of a zero-sum superbase form a primitive basis, so both matrices are unimodular and $\mathbf{P} = \mathbf{U}(A)\mathbf{U}(B)^{-1}$ is an exact integer matrix. It is accepted if $\mathbf{P}^{\mathsf{T}}\mathbf{G}(A)\mathbf{P} = \mathbf{G}(B)$ to within a residual tolerance $\max(\texttt{verify_abs},\ \texttt{verify_rel} \cdot \operatorname{tr}|\mathbf{G}(B)|)$. The residual is linear in the metric, so its noise model is Gaussian with scale σ_p; no square root is involved at this stage. Equivalently, if the rows of A are the integer coordinates of three superbase vectors of R in $\mathbf{B}_R$ and the rows of B those of the corresponding vectors of N in $\mathbf{B}_N$, the two vector triples coincide in space exactly when $\mathbf{B}_R A^{\mathsf{T}} = \mathbf{B}_N B^{\mathsf{T}}$, so $\mathbf{P} = A^{\mathsf{T}} B^{-\mathsf{T}}$ with both factors unimodular.

D.2. The procedure as pseudo-code

```
PREPARE_REFERENCE(R_conv, sigma_theta):       # run once, cache the result
  C_R, G_R <- centering matrix and primitive metric of R   # rational C_R
  U_R      <- selling_reduce(G_R)   # 4x3 integer rows, obtuse superbase
  tau      <- 3 * sigma_theta * T(G_R) / 6      # invariant zero tolerance
  S_R      <- closure(U_R, G_R, tau)  # ALL obtuse superbases, <= 32
  key_R    <- sort(sqrt(conorms(U_R, G_R)));  radius <- from sigma (App. C)
  H        <- REINDEX_FRAME(ref, R_conv)   # automorphism coset, = Aut(R)
  if space group known: L <- Laue group;
      branches <- one representative per coset of L in H   (exact H only)
  return ref = (C_R, G_R, U_R, S_R, key_R, radius, H)

REINDEX_FRAME(ref, N_conv, sigma_theta):      # per frame
  C_N, G_N <- centering matrix and primitive metric of N
  if |key(G_N) - ref.key_R| > ref.radius: return EMPTY   # optional screen
  U_N <- selling_reduce(G_N)
  S_N <- closure(U_N, G_N, tau)   # usually {U_N}; larger near a stratum
  ops <- {}
  for C_R' in ref.S_R, C_N' in S_N, pi in S4, eps in {+1,-1}:
      A <- rows 1..3 of eps*C_R'[pi];  B <- rows 1..3 of C_N'
      P <- A^T B^{-T}                 # Bprim_N = Bprim_R P ; integer, |det|=1
      if || P^T ref.G_R P - G_N || <= tol(sigma): ops <- ops + {P}
  ops <- deduplicate(ops)             # = P0 * ref.H ; report (P0, H), by det
  if ref.branches: report (P0, ref.branches)   # |H|/|L| alternatives
  # x_R = P x_N ; h_R = P^{-T} h_N ; W_R = P W_N P^{-1} ; P_conv = C_R P C_N^{-1}
  return ops
```

D.3. Pseudo-symmetry by closure cosets

```
PSEUDO_SYMMETRY(ref, delta):           # once per reference; delta in conorm units
  cf <- coform(ref.U_R, ref.G_R)       # 2x3 conorms, integer coords retained
  H  <- ref.H                          # exact Aut(R) from the closure self-match
  patterns <- degeneration patterns of cf:      # finite list
      zero sets   {p_ij : p_ij < delta}
      equal sets  {(p_ij, p_kl) : |p_ij - p_kl| < delta}, extended to
                  Aut(R)-orbits, and their unions (closed under Aut(R))
  out <- {}
  for pat in patterns:
      cf* <- impose(pat, cf)           # zero / average whole orbits
      d   <- |cf - cf*|;  if d > delta: continue
      G*  <- Gram from cf* (Kurlin Lemma 6.2), same superbase coordinates
      S*  <- closure(ref.U_R, G*, tau);  H* <- self-match of G* over S*
      assert H subset H*
      for P in H* \ H: residual(P) <- |P^T ref.G_R P - ref.G_R|
      out <- out + {(pat, d, cosets H*/H, cosets H*/L, residuals)}
  return out                           # cached with ref
```

The imposition step must act on whole Aut(R)-orbits of coform entries; otherwise R^* depends on which member of the orbit was edited and $H \subseteq H^*$ can fail.[S17] The checks reported with this paper do not execute the pattern loop. They impose two strata by hand, as stated in Section 2.8: $a = c$ on the monoclinic family, and equal

short edges on the near-tetragonal cuboid.

D.4. Relation to the Le Page route

Le Page's algorithm finds lattice directions $[uvw]$ and planes (hkl) that are parallel to within δ, that is, approximate twofold axes of $\mathbf{G}$, and assembles the metric point group from them; reindexing operators are then taken from that group, and two settings are related only if the relating operator happens to be a member. This is the right construction for pseudo-symmetry and is unsuited to recovering the operator relating two settings: that operator is exact for every distortion, but its twofold residual grows linearly with the distortion and leaves the gate (Table 3). The closure procedure asks whether $\mathbf{P}$ carries $\mathbf{G}(R)$ to $\mathbf{G}(N)$ rather than whether $\mathbf{P}$ is a near-symmetry of $\mathbf{G}(R)$, and it is complete because every obtuse superbase of each lattice is on the list. The exact automorphism group is obtained as the self-match; near-automorphisms, when wanted, remain a tolerance-based search for which the twofold method is appropriate.

D.5. Relation to Lebedev's coset reduction

Lebedev *et al.* (2006) reduce the operators that intensity statistics must examine to coset representatives of the Laue group L in the lattice symmetry group H. For index 2 the non-trivial coset always contains a twofold; for higher index it need not (the coset $\{4, 4^3\}$ of 2 in 422 contains none), so representatives are chosen of minimal order.[S16] Their H is obtained from a tolerance search on the metric, which is appropriate for the pseudo-merohedral case they address. Here $H = \mathrm{Aut}(R)$ is obtained exactly by matching the reference against itself over its cached closure, and the coset argument applied to that exact H reduces the ambiguity handed to intensities from $|H|$ operators to $|H|/|L|$ branches, computed once per reference. If pseudo-merohedral branches are wanted, the same coset argument is applied to the tolerance-enlarged H^* of Section 2.8

in a separate, declared step.

D.6. Completeness of the closure route versus a bounded unimodular search

The closure route is complete because every obtuse superbase of each lattice is enumerated (Kurlin, 2026, Lemmas 4.1–4.5) and the reduction bookkeeping absorbs whatever large integers the input bases carried; no bound on the entries of $\mathbf{P}$ is assumed, and the large entries that genuine cosets contain are produced rather than searched for. What a search over unimodular matrices with entries in $\{-1, 0, 1\}$ can and cannot do between reduced settings is settled by the following machine-checked proposition.

Proposition. (i) On each Voronoi stratum the closure, expressed in the integer coordinates of one obtuse superbase, is a fixed finite set of superbases — 1, 2, 3, 6 and 16 classes modulo relabelling and sign for V1 through V5 — each obtuse on the entire open stratum, because every conorm of every member is a nonnegative integer combination of the stratum's free conorms. (ii) Between any two bases drawn from these superbases, the relating coset $\tilde{B}^{-1}\,\mathrm{Aut}(\mathbf{G})\,\tilde{A}$ contains an operator with entries in $\{-1, 0, 1\}$. (iii) It is not contained in them: in V4, 18 of the 24 reduced bases have a self-coset operator with an entry of two; in V5, 60 of 64, and bases from different arrangement classes are related by coset operators with an entry of three. *Proof.* The metric tensor is linear in the conorms, so whether a slot of a candidate superbase vanishes identically on the stratum is read off an integer coefficient vector, not off parameter values; the flip graph grown from the standard superbase by flips at identically-zero slots is therefore the same for every lattice of the type, and one breadth-first enumeration per type covers them all. Signed permutations of either basis multiply the relating operators on the outside and cannot change entry magnitudes, and a larger point group at a special lattice only enlarges the coset, so (ii) extends from the generic enumeration to every lattice. The finitely many operators

are then listed and their entries read off.[S22]

In V1 through V3 the enumeration shows every coset operator staying inside $\{-1, 0, 1\}$: the coset escape is a V4 and V5 phenomenon, driven by skew member bases conjugating the point group. A $\{-1, 0, 1\}$ search between Selling-reduced settings therefore always finds an operator and never the guarantee of the full coset. Between Niggli cells of one lattice the corresponding entry bound holds empirically for the whole coset (exact or noisy, any input setting, including adversarial anisotropic and near-degenerate cases),[S22] because the Niggli basis is unique and axis-aligned, so the coset is the point group up to signed permutations. It stays empirical for a structural reason: Niggli reduction has no closure equivalent — its case structure is the parameter-dependent chamber decomposition of Gruber (1973) and Andrews & Bernstein (2014), not a stratum-fixed integer object, so no single finite enumeration settles it, and shortness alone cannot substitute, since the cuboid example above shows bases built entirely from short vectors related by an entry of three. A proof would have to excavate those chambers as polyhedral cones in conorm space; the enumeration here supplies the finite skeleton for that and we leave it as future work. Niggli-first therefore makes the small search coset-complete in practice, at the price of reintroducing the representative's boundary behaviour as an implementation dependency and of testing 6960 candidates where the closure needs at most 32. The implementation of the Niggli change-of-basis must itself be verified: the returned matrix N should satisfy $N^{\mathsf{T}}\mathbf{G}N = \mathbf{G}_{\mathrm{red}}$ exactly, and this identity belongs in the test suite.[S22] Between different lattices no same-lattice proposition applies and equation (2) is the only guarantee; on a Selling-reduced needle pair with certificate radius $R \approx 7$, the radius-two ball improves the radius-one minimum by a factor of 2.8, so the truncation there returns a wrong ideal, exactly as the certificate warned.[S22]

D.7. If nothing is accepted

Either N is a different lattice; or N is a sub- or supercell of R (a different lattice, requiring finite-index enumeration outside this procedure); or the tolerance is tighter than the noise. The residual of the best-scoring candidate is reported in all cases, so the third possibility is visible rather than silent.

D.8. The closure match is a congruence test, not a metric

The minimum aligned distance over $\mathcal{S}(A) \times \mathcal{S}(B)$ is exact under the congruence hypothesis and, between genuinely different lattices, is an upper bound on any frame infimum — and not a tight one: it is discontinuous across Selling faces. A pair constructed to converge through a face, one lattice with a conorm at $+\delta$ and the other realized from the same frame with that conorm at $-\delta$, has key distance of order δ, while the closure minimum stays near a constant 0.51 as $\delta \to 0$: an overestimate by factors 3, 14 and 71 at $\delta = 0.5$, 0.1 and 0.02, because the two reduced forms land in different arrangements (the face flip is not a relabelling) and the test pays the full inter-arrangement separation.[S31] On deposited cells the same effect appears on the Hfq pair 4nl7→2yht: key and closure minimum agree at 5.43 Å, while admitting frames with conorms down to a declared fraction of the scale lowers the match to 2.83 and then 2.65 Å — the same verdict NCDist reaches for this pair, whose value is on an arbitrary internal scale and is not compared by magnitude (McCoy *et al.*, 2026). The key does not bound these widened values, since the sorted conorms of a non-obtuse frame are not the key.[S31] One case is exactly solvable: if the closure minimum is smaller than every strictly positive normalised conorm of both lattices, no frame within that distance of either can cross a face, widening cannot improve the match, and the closure minimum is the frame infimum; in 40 random interior pairs, widening improved the minimum only where a lattice margin lay below it.[S31] For similarity

between different lattices, then, the key is the coordinate, the widened match is an explicitly labelled deformation question, and the closure minimum is reserved for the hypothesis it was built for.

D.9. Guarantee under the noise model

Under the congruence hypothesis the chain of Section 2.4 is provable step by step. *Retrieval*: the linear key is Lipschitz in the metric tensor, so two measurements of one lattice have key distance of order η, and a radius at a stated multiple of η retrieves the pair; Lemma 1 makes the radius sufficient as well as necessary. *Coverage*: Selling reduction of a measured cell applies integer moves triggered by measured conorms, and every slot it leaves non-negative is, in the true lattice, at worst $-\eta$; the reduced measured frame is therefore a presentation of the truth obtuse to within η, and every member of the true closure is reached from it by flips at slots whose true conorms vanish and whose measured values therefore lie within a small multiple of η of zero — with the zero tolerance at that multiple, every needed flip triggers, and the widened closure of each measured cell contains a counterpart of the entire true closure, with integer provenance.[S32] *Coset*: for each true operator the matched counterparts are present, the operator's entries are bounded over the closure, and its residual on the measured metrics is $\|\mathbf{P}^{\mathsf{T}} E_A \mathbf{P} - E_B\|$, of order η; a verification residual at the stated multiple accepts the full coset, and the partial coset of Section 4.3 is exactly what a residual below it produces. *Soundness*: acceptance requires an exact integer map carrying one measured metric onto the other within the residual, and with exact data only true operators satisfy it.[S32] The constants are small and computable, and the Supporting Information verifies the chain across the five Voronoi types together with its failure when a tolerance is set below its multiple.

Appendix D

Data and code availability

The cell key, lattice comparison, reindexing coset and PDB radius search are library calls in `agentsg.cell` and, unchanged, on a local read-only service (`agentsg.serve`), whose REST endpoints are `GET /v1/cell`, `POST /v1/compare`, `POST /v1/reindex` and `POST /v1/pdb/search`; the same four calls are exposed as MCP tools. Space-group, reflection, site and plate tools served by the same process are outside the scope of this paper. The source is `https://github.com/phzwart/agentsg`. An archival DOI for that repository has not been minted. The 206 214 PDB cells used for the index are the snapshot `https://doi.org/10.5281/zenodo.22986222`. Lysozyme cells derive from public PDB entries (UniProt P00698). Serial data are CXIDB entry 83 (DOI 10.11577/1464102), the β-lactamase stream of Wiedorn *et al.* (2018). The Supporting Information lists, for every marked statement, the computation that checks it.

This work was carried out in the Molecular Biophysics and Integrated Bioimaging Division, Lawrence Berkeley National Laboratory. No grant or contract number is recorded in the project sources.

References

Andrews, L. C. & Bernstein, H. J. (1988). *Acta Cryst.* A**44**, 1009–1018.

Andrews, L. C. & Bernstein, H. J. (2014). *J. Appl. Cryst.* **47**, 346–359.

Andrews, L. C. & Bernstein, H. J. (2016). *J. Appl. Cryst.* **49**, 756–761.

Andrews, L. C. & Bernstein, H. J. (2023). *Acta Cryst.* A**79**, 485–498.

Andrews, L. C., Bernstein, H. J. & Sauter, N. K. (2019). *Acta Cryst.* A**75**, 593–599.

Bravais, A. (1850). *J. Éc. Polytech.* **19**, 1–128.

Brehm, W. & Diederichs, K. (2014). *Acta Cryst.* D**70**, 101–109.

Bright, M., Cooper, A. I. & Kurlin, V. (2021). arXiv:2109.11538.

Conway, J. H. & Sloane, N. J. A. (1992). *Proc. R. Soc. London Ser. A*, **436**, 55–68.

Delone, B. N. (1932). *Z. Kristallogr.* **84**, 109–149.

Fiedler, M. (1973). *Czech. Math. J.* **23**, 298–305.

Gildea, R. J. & Winter, G. (2018). *Acta Cryst.* D**74**, 405–410.

Grosse-Kunstleve, R. W., Sauter, N. K. & Adams, P. D. (2004). *Acta Cryst.* A**60**, 1–6.

Gruber, B. (1973). *Acta Cryst.* A**29**, 433–440.

Kurlin, V. (2026). *A complete isometry classification of 3-dimensional lattices.* Revised manuscript, April 2026. https://kurlin.org/projects/lattice-geometry/lattices3Dmaths.pdf.

Lebedev, A. A., Vagin, A. A. & Murshudov, G. N. (2006). *Acta Cryst.* D**62**, 83–95.

Le Page, Y. (1982). *J. Appl. Cryst.* **15**, 255–259.

McCoy, A. J., Andrews, L. C., Bernstein, H. J. & Read, R. J. (2026). *Acta Cryst.* D**82**, 813–823.

McGill, K. J., Asadi, M., Karakasheva, M. T., Andrews, L. C. & Bernstein, H. J. (2014). *J. Appl. Cryst.* **47**, 360–364.

Niggli, P. (1928). *Krystallographische und strukturtheoretische Grundbegriffe.* Handbuch der Experimentalphysik, Vol. 7, Part 1. Leipzig: Akademische Verlagsgesellschaft.

Wiedorn, M. O. *et al.* (2018). *Nature Commun.* **9**, 4025.

Table 1. *The Selling closure by Voronoi type (Kurlin, 2026, Lemmas 4.1–4.5). "Classes" are families of obtuse superbases not related by relabelling; the group in parentheses is the relabelling ambiguity remaining inside the coform once the zero pattern is fixed: S_4 on all six conorms (V1), D_4 on the 2×2 nonzero block (V2), S_4 on the four nonzero conorms (V3), S_3 on three conorms with one fixed (V4), and the three nonzero conorms free (V5). "Zeros" is the number of zero conorms in the reduced superbase, "fibre" the generic number of non-isometric lattices sharing one sorted key. Superbases are counted as unordered sets of four vectors, so a generic lattice contributes two (the superbase and its global negation).*

type	Voronoi cell	zeros	superbases	classes (group)	fibre
V1	truncated octahedron	0	2	1 (S_4)	30
V2	hexa-rhombic dodecahedron	1	4	2 (D_4)	15
V3	rhombic dodecahedron	2, opposite	6	3 (S_4)	1
V4	hexagonal prism	2, at a vertex	12	3 (S_3)	4
V5	cuboid	3	32	4 (free)	1

Table 2. *Four sources of ambiguity in assigning indices, and where each is resolved.*

ambiguity	what it is	resolved by	tolerance
setting	same lattice, different cell (bookkeeping)	closure and metric test: unique $\mathbf{P}_0$ modulo H	noise residual only
exact symmetry	$H = \mathrm{Aut}(R)$ modulo Laue group L	intensities, or carried as H/L	none
pseudo-symmetry	automorphisms of a nearby lattice, $H^* \supset H$	Section 2.8; intensities over H^*/L	one declared distance
key collision	different lattices, same sorted key	certification stage	none

Table 3. *Monoclinic axis exchange under increasing distortion. The two settings describe the same lattice throughout. A 3° pseudo-symmetry gate eventually drops the exchange; closure-based recovery returns the exact integer operator at every distortion.*

$\lvert a - c \rvert$ (Å)	flip δ (°)	key dist.	δ-gate	exact op.	residual
0.0	0.00	0	keeps	yes	0
1.0	1.04	7×10^{-15}	keeps	yes	0
2.0	2.09	2×10^{-14}	keeps	yes	0
2.85	2.98	7×10^{-15}	keeps	yes	0
4.0	4.18	1×10^{-14}	drops	yes	0
5.0	5.22	2×10^{-14}	drops	yes	0
10.0	10.41	7×10^{-15}	drops	yes	0

Table 4. *The Bravais skeleton of the volume-normalised key space: fixed points, exact nexus and asymptotic seams. Coordinates and distances are dimensionless (key over $V^{1/3}$).*

object	where	value
cP point	terminus of tP, oP, oC and hR(α=90°)	$(0, 0, 0, 0.577{\times}3)$
cI point	hR(α=109.47°) = tI(c/a=1); oI limit	$(0.257{\times}6)$
cF point	hR(α=60°) = tI(c/a=$\sqrt{2}$); oF limit	$(0, 0, 0.397{\times}4)$
oC through tP	oC(a=b) coincides with tP	exact
oI, oF feeds	oI(a=b) = tI; oF(a=b) = tI($\sqrt{2}\,c/a$)	exact
seam hP ∥ hR	hP(R) beside hR(hex 3R)	0.093 → 0.0009, R=1 → 32
seam tP ∥ tI	tP(R) beside tI(2R); flat: tI($R/\sqrt{2}$)	0.64 → 0.004, R=1 → 64
curve separation	cP to the tI curve; cI to the tP curve	0.50; 0.71

Table 5. *The ladder: the ideal deformation distance and its certified accelerations.*

object	relation to the ideal	guarantee
ideal $d(A, B)$, eq. (1)	itself	value D bounds $\|\mathbf{M}\|$; one pass certifies
widened minimum	frames with penetration $\leq \tau$	decreases to the ideal; certified stop
closure minimum	jointly obtuse frames	= ideal under the margin certificate
congruence test	the ideal thresholded at noise	complete under the noise model
sorted key	retrieval rung	exact at the zero set; conservative radius
radius-one ball (6960)	truncated enumeration	correct iff certificate radius < 2; never coset-complete for V4, V5
Le Page search	angular proxy of the self-coset	none; subsumed

Table 6. *Median key shift (Å) under Gaussian angular noise on the $P6_3$ cell* $(41.8, 41.8, 233, 90, 90, 120)$*; invariant floor $s = \sigma_\theta T$, soft threshold $\kappa = 2$, linear key $p/\sqrt{T}$.*

σ_θ	$\sqrt{p}$	$\sqrt{p+s} - \sqrt{s}$	$\sqrt{\max(p - 2s, 0)}$	$p/\sqrt{T}$
0.01°	1.58	0.28	0.03	0.01
0.05°	3.12	0.64	0.17	0.06
0.10°	4.63	0.68	0.35	0.10

Table 7. *Anisotropy budget of the Niggli-reduced Scotty cells (McCoy* et al.*, 2026): the radius-one search is certified for errors $D < D_{\max} = 4\lambda_{\min} - \lambda_{\max}$ of the Niggli metric tensor; a negative budget means it runs uncertified at every error level. Seven of seventeen are negative.*

cell	case	$\lambda_{\max}/\lambda_{\min}$	$R(0)$	$D_{\max}$ (Å^2)
4amt	prorenin	3.8	1.94	912
5mlg	prorenin	3.8	1.94	960
5mkt	prorenin	3.7	1.93	1141
2d4k	lysozyme	5.5	2.34	−1077
1vdp	lysozyme	5.0	2.24	−808
1fe5	PLA_2	2.9	1.71	1217
1u4j	PLA_2	1.0	1.01	9619
1g2x	PLA_2	1.0	1.01	9694
7mg1	concanavalin A	3.9	1.98	148
1hqw	concanavalin A	4.0	1.99	42
5wey	concanavalin A	4.1	2.02	−152
4nl6	GabD contaminant	4.4	2.10	−1541
1jr7	GabD contaminant	4.4	2.11	−1739
2r6s	GabD contaminant	4.5	2.11	−1763
8cdf	GabD contaminant	4.4	2.10	−1644
4nl7	Hfq contaminant	3.0	1.73	1955
2yht	Hfq contaminant	3.1	1.75	1739

Synopsis

A unit cell is one description of a lattice, not the lattice itself. Sorting a lattice's normalised Selling scalars gives every basis the same continuous Euclidean coordinate, whose distance never exceeds the true lattice distance; on it the Protein Data Bank becomes an atlas of lattice shapes organised by a computable Bravais skeleton, and the finite set of obtuse superbases returns exact reindexing operators with their full symmetry cosets.